Venkata Lakshmi Parimala Pasupuleti

AI Platform Engineer · Python / LangGraph / MCP · Agentic Architecture · LLMOps & Observability · Platform Governance & Responsible AI

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SUMMARY

AI Platform Engineer with 9 years of production software ownership, hands-on agentic architecture, and enterprise platform governance experience. Built AgentProbe -- an open-source LLMOps platform with AI observability infrastructure (agent tracing, logs/metrics/traces, CI regression gate, LLM-as-judge, hallucination detection), multi-provider LLM routing (Anthropic/OpenAI/Groq/Ollama), and Responsible AI guardrails (filters, rate limiting, PII/PHI controls, output validation). Designed and delivered production retrieval-augmented generation (RAG) pipelines and multi-agent LangGraph systems -- implementing Model Context Protocol (MCP) integration and agentic workflow orchestration with memory layers and state management. Seven years at Entrata (PropTech SaaS, 100K+ MAU) owning HIPAA-compliant data pipelines, RBAC access governance, and containerized microservices at enterprise scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI Platform & Agentic Architecture: Python, LangGraph, LangChain, Agentic AI, Agentic Workflows, Multi-Agent Orchestration, MCP (Model Context Protocol), Tool Calling, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, HITL, Memory Layers, State Management, LLM Routing, Multi-Provider LLM Gateway (Anthropic, OpenAI, Groq, Ollama), Prompt Engineering, ReAct

LLMOps & Platform Governance: AI Observability, Agent Tracing, Hallucination Detection, LLM Evaluation, LLM-as-Judge, CI/CD (GitHub Actions), Responsible AI, Guardrails, Rate Limiting, PII/PHI Handling, RBAC, Access Controls, Least Privilege, Identity Governance, Policy Enforcement, Red-Teaming, Output Validation, Configuration Management

Infrastructure & Full-Stack: Python, JavaScript, TypeScript, SQL, FastAPI, Flask, REST APIs, Microservices, Data Pipelines, ETL, PostgreSQL, AWS (S3, EC2, RDS, Glue, Lambda, IAM), Azure, GCP, Docker, Kubernetes (K8s), Git, Linux, Agile/Scrum, Fine-tuning (HuggingFace, PyTorch), NLP, BERT, spaCy

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI Engineer (Customer-Facing Delivery)

- Designed and delivered a production LangGraph agentic AI system for an external enterprise client -- full technical ownership: LLM selection, RAG pipeline over a regulatory corpus, prompt engineering strategy, output validation, and integration with enterprise data sources via MCP-compatible API connectors.
- Built Responsible AI validation guardrails enforcing compliance rules deterministically (filters, rate limiting, PII controls); shipped eval harness alongside the product so every output was validated before production delivery.
- Facilitated design reviews to align solution architecture with stakeholder requirements; iterated on agentic workflows and automated evaluations in an agile cadence.

Entrata (PropTech SaaS · 100K+ MAU · HIPAA/PHI · enterprise data pipelines)

Sep 2018 - Aug 2024

Senior Software Engineer

- Owned HIPAA-compliant data pipelines for 7 years (180,000 HUD records, 100K+ units) with RBAC access governance, least-privilege data access controls, and PHI handling -- zero compliance violations across 7 years of production operation.
- Shipped 20+ full-stack production modules across 120+ Agile sprints; led Docker containerization of microservices (50% scalability improvement, 30% deployment time reduction); maintained CI/CD pipeline hygiene and change management standards.
- Led AWS S3 document migration -- IAM roles, Libsodium-encrypted pipelines, 100% PHI compliance, zero security incidents; built Python/Bash automation eliminating 80% of manual operational workload and reducing runbook execution time.
- Optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers; 300+ peer code reviews focused on security, reliability, and maintainability.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Fine-tuned a custom classification model using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented model design and evaluation results to faculty.
- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest anomaly detection across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: AI Platform, LLMOps & Observability Infrastructure

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · LangChain · ChromaDB · MCP (Model Context Protocol) · CI/CD (GitHub Actions) · Multi-provider LLM gateway (Anthropic/Claude, OpenAI, Groq, Ollama)

- Open-source AI platform with production LLMOps and full observability: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, agent tracing (logs/metrics/traces), multi-provider LLM gateway with rate limiting and spend controls, and Responsible AI guardrails (filters, red-teaming, PII controls, output validation). Built to standardize how agentic AI systems are evaluated, governed, and deployed before they touch production users -- the operational backbone for safe AI adoption.

Leasel: Agentic RAG System with HITL & Access Governance

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit · Responsible AI

- Production agentic AI platform: 8 LangGraph agents with retrieval-augmented generation, memory layers, state management, HITL access governance (regulated decisions always route to human -- safety boundary enforced in code), and conditional routing. 93.6% precision, 87.9% F1 across 1,240 test cases. PropTech / real estate domain (lease compliance, affordable housing regulations, property data) -- direct Realtor.com industry alignment.

AI on FHIR: HIPAA-Compliant Healthcare AI with RBAC & PII Controls

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR APIs · spaCy · BERT · Docker · HIPAA · RBAC · Least Privilege

- HIPAA-compliant AI system over structured PHI -- RBAC, least-privilege access controls, PII handling, identity governance, NLP-to-FHIR API translation (EN/ES), LLM + BERT integration; reduced clinical retrieval time ~60% with zero compliance incidents.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017